

of bandwidths, which we recommend using in practice while aggregating over both Gaussian and Laplace kernels, each with multiple bandwidths. This adaptive collection is the discretisation of an interval and in practice we found that using ten bandwidths per kernel performs as well as using the whole continuous interval in the limit. Our MMDAgg test obtains significantly higher power than other state-of-the-art MMD-based two-sample tests in synthetic settings where the smoothness Sobolev assumption is satisfied, this empirically validates our theoretical result of minimax optimality and adaptivity over Sobolev balls. In experiments on image data, we observe that MMDAgg almost matches the power of much more complex two-sample tests relying on training models, such as neural networks, to detect the difference in distributions.

We now discuss three research directions based on this current work, two of which have been explored by Schrab et al. (2022a,b).

First, it would be interesting to consider the two-sample kernel-based test of Jitkrittum et al. (2016), who use adaptive features (in the data space or in the Fourier domain) to construct a linear-time test with good test power. Jitkrittum et al. (2016) require setting aside part of the data to select the kernel bandwidths and the feature locations, by maximizing a proxy for test power. They then perform the test on the remaining data. It would be of interest to develop an approach to learning such adaptive interpretable features without data splitting. Adapting the current results of this work to that setting remain an open challenge.

Second, aggregated tests that are adaptive over Sobolev balls have been constructed for several alternative testing scenarios. The independence testing problem using the Hilbert Schmidt Independence Criterion has been treated by Albert et al. (2022), which is related to the Maximum Mean Discrepancy (Gretton et al., 2012a, Section 7.4). A further setting of interest is goodness-of-fit testing, where a sample is compared against a model. Our theoretical results can directly be applied to goodness-of-fit testing using the MMD, as long as the expectation of the kernel under the model can be computed in closed form. A more challenging problem arises when this expectation cannot be easily computed. In this case, a test may be constructed based on the Kernelised Stein Discrepancy (KSD—Liu et al., 2016; Chwialkowski et al., 2016). This corresponds to computing a Maximum Mean Discrepancy in a modified Reproducing Kernel Hilbert Space, consisting of functions which have zero expectation under the model. Building on the present paper, Schrab et al. (2022a) develop an adaptive aggregated KSDAgg test of goodness-of-fit for the KSD and provide conditions which guarantee high test power for KSDAgg.

Third, our MMDAgg test proposed in this work, the KSDAgg test of Schrab et al. (2022a), and the aggregated HSIC test of Albert et al. (2022), are all quadratic-time hypothesis tests. While quadratic-time tests usually achieve higher power than linear-time tests, this comes at the expense of an important computational cost. To tackle this problem, relying on incomplete U -statistics, Schrab et al. (2022b) propose efficient variants (including linear-time ones) of those three aggregated tests, called MMDAggInc, KSDAggInc and HSICAggInc. They theoretically quantify the cost incurred in the minimax rate over Sobolev balls for this improvement in computational efficiency.

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Overview of Appendices

In Appendix A, we present results of additional experiments. In Appendix B, we explain the relation between using permutations and using the wild bootstrap. We present an efficient implementation of MMDagg in Appendix C. We highlight the proof strategy of deriving the minimax rate over a Sobolev ball in Appendix D. Finally, we present the proofs of all our results in Appendix E.

Appendix A. Additional experiments

In this section, we verify the level achieved by all the tests considered (Appendix A.1), and run multiple power experiments: widening the collection of bandwidths (Appendix A.2), comparing wild bootstrap and permutations (Appendix A.3), using unbalanced sample sizes (Appendix A.4), and increasing the sample sizes for the `ost` test (Appendix A.5).

A.1 Level experiments

In Table 2, we empirically verify that all the tests we consider have the desired level $\alpha = 0.05$ in the three different settings considered in Figures 3 to 5. For the aggregated MMDagg tests of Section 5.1, we use the collection of bandwidths $\Lambda(-4, 0)$ for samples drawn from a uniform distribution in one and two dimensions. For samples drawn from the set $\mathcal{P}$ of images of all MNIST digits, we use $\Lambda(10, 14)$ and $\Lambda(12, 16)$ for the Gaussian and Laplace kernels, respectively. To obtain more precise results, we use 5000 repetitions to estimate the levels.

We observe in Table 2 that all the tests have well-calibrated levels, indeed all the estimated levels are relatively close to the prescribed level 0.05. We consider three different types of data and run the tests with the Gaussian and Laplace kernel using either a wild bootstrap or permutations. We note that there is no noticeable trend in the differences in the estimated levels across all those different settings.

Table 2: Level experiments with samples drawn either from d -dimensional uniform distributions or from the MNIST dataset using the Gaussian (G.) and Laplace (L.) kernels with either a wild bootstrap (w.b.) or permutations (p.).

			MMDAgg uniform	MMDAgg centred	MMDAgg increasing	MMDAgg decreasing	median	split	ost
$d = 1$	G.	w.b.	0.0476	0.052	0.0456	0.0434	0.047	0.054	0.0594
		p.	0.0496	0.0532	0.0478	0.0454	0.0468	0.0528	0.0594
	L.	w.b.	0.0474	0.0488	0.0516	0.0504	0.0534	0.05	0.0586
		p.	0.047	0.0482	0.0496	0.0494	0.0522	0.0494	0.0586
$d = 2$	G.	w.b.	0.039	0.0432	0.044	0.0496	0.0464	0.0482	0.0478
		p.	0.0424	0.0446	0.0414	0.0498	0.0466	0.0472	0.0478
	L.	w.b.	0.0382	0.0502	0.0506	0.0478	0.0438	0.0548	0.0502
		p.	0.0418	0.0474	0.0514	0.049	0.0458	0.0548	0.0502
MNIST	G.	w.b.	0.0478	0.0528	0.0474	0.0488	0.0526	0.0498	0.0496
		p.	0.042	0.05	0.0476	0.048	0.055	0.0484	0.0496
	L.	w.b.	0.054	0.052	0.0424	0.0548	0.0518	0.0444	0.05
		p.	0.0526	0.0532	0.0442	0.0554	0.051	0.0448	0.05

		MMDAgg [*] Laplace	MMDAgg [*] Gaussian	MMDAgg [*] Laplace & Gaussian	MMDAgg [*] All	AutoML
$d = 1$	w.b.	0.0506	0.05	0.0496	0.0492	0.0462
	p.	0.0528	0.0492	0.0496	0.0494	
$d = 2$	w.b.	0.0458	0.0428	0.0434	0.043	0.051
	p.	0.0456	0.0438	0.0438	0.0434	
MNIST	w.b.	0.0624	0.0624	0.0632	0.0594	0.518
	p.	0.0612	0.062	0.0628	0.0622	

A.2 Power experiments: widening the collection of bandwidths

In practice, we might not have strong prior knowledge to guide us in the choice of a collection consisting of only a few bandwidths. For this reason, in practice, we recommend using the adaptive parameter-free collection introduced in Section 5.2 with both Laplace and Gaussian kernels. Nonetheless, it is interesting to study the properties of the family of collections of Section 5.1 consisting of the median bandwidth scaled by powers of 2. In Figure 7, we design an experiment where we start with a collection of 3 bandwidths chosen to be centred around the optimal bandwidth. We then widen the collection of bandwidths and observe how much the power deteriorates as more bandwidths are included in the collection.

We consider collections ranging from 3 to 15 bandwidths for our two tests **MMDAgg uniform** and **MMDAgg centred**, for the three types of data used in Figures 3 to 5. For the perturbed uniform distributions in one and two dimensions, we use the collection of bandwidths $\Lambda(-2-i, -2+i)$ for $i = 1, \dots, 7$ for both kernels. For the MNIST dataset with